

CERTIFIED ETHICAL HACKER · FINAL COURSE PROJECT

PROJECT 3

Multi-Router OSPF Network

Design · Configuration · Verification · Hardening

OSPFv2 · Single Area 0 · Redundant Ring · MD5 Authentication

Platform	Cisco Packet Tracer
Devices	4 × Cisco 2811 Routers · 4 × Cisco 2960 Switches · 8 × PCs
Routing Protocol	OSPFv2 (Open Shortest Path First), Backbone Area 0
Security	OSPF MD5 Message-Digest Authentication
Prepared by	Anwar

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1 Executive Summary

This report documents the design, configuration, verification, and hardening of a four-router Open Shortest Path First (OSPF) network built in Cisco Packet Tracer. Four Cisco 2811 routers are interconnected in a redundant ring using point-to-point serial links, with each router serving a dedicated LAN of two end hosts through a Cisco 2960 switch. A single OSPF backbone area (Area 0) distributes routing information across the entire topology.

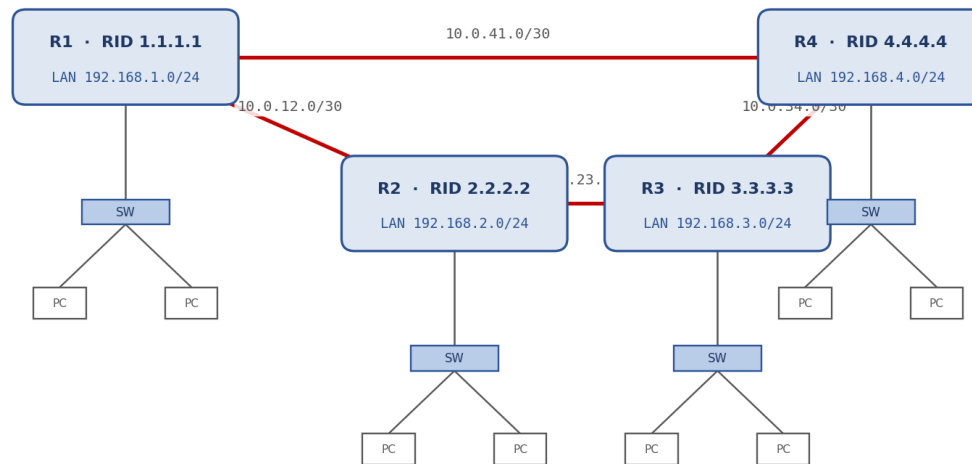
All adjacencies were verified in the FULL state, every router learned the complete network via OSPF, and end-to-end connectivity across the ring was confirmed with a zero-loss ping and a traceroute. As a Certified Ethical Hacker exercise, the routing control plane was then hardened with MD5 neighbor authentication to defend against spoofed-update (LSA injection) attacks; authentication was confirmed enabled while adjacencies remained FULL.

2 Objectives

- Design a four-router ring topology using Cisco 2811 routers and WIC-2T serial modules.
- Assign a structured IP addressing scheme using /30 serial links and /24 LANs.
- Configure OSPFv2 in a single backbone area (Area 0) on all four routers.
- Verify full adjacency, complete route distribution, and end-to-end reachability.
- Demonstrate the redundancy provided by the ring topology (equal-cost dual paths).
- Harden the OSPF control plane using MD5 authentication (security objective).

3 Network Topology

The four routers form a closed ring: R1–R2–R3–R4–R1. Each router connects to its two neighbors through serial interfaces and to its local LAN through FastEthernet0/0 and a 2960 switch. Figure 1 is the logical design; Figure 2 shows the working topology in Packet Tracer.

Figure 1. Four-Router OSPF Ring Topology (Area 0)

All serial links are /30 point-to-point in OSPF area 0. Each router hosts a LAN via a 2960 switch.

Figure 1. Logical OSPF ring topology with per-link /30 serial subnets and per-site /24 LANs.

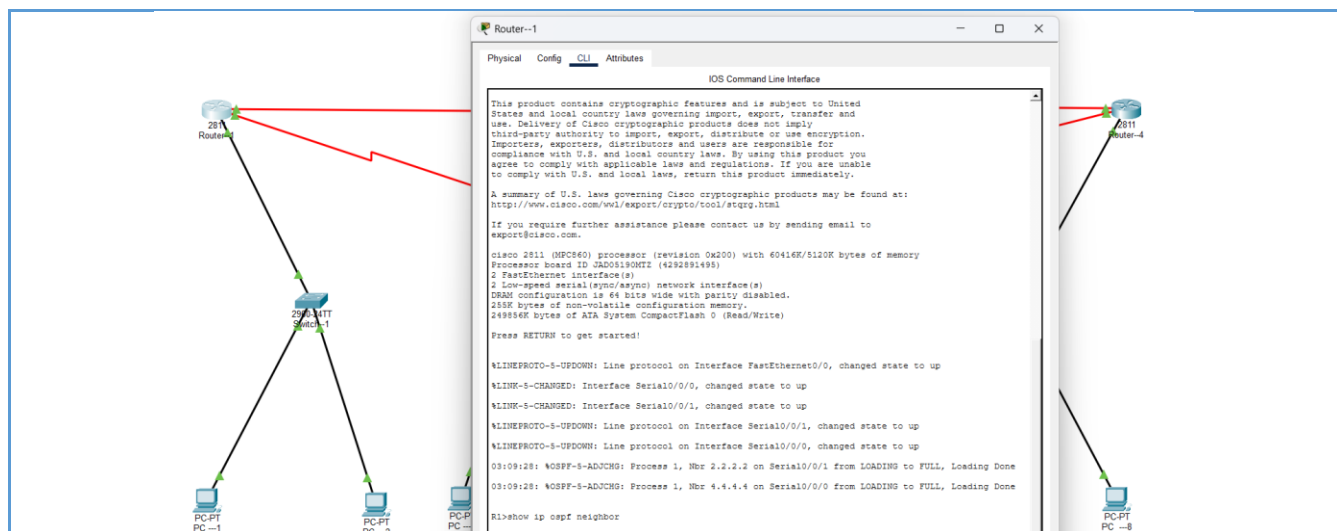


Figure 2. Live Packet Tracer topology — four routers in a closed serial ring, each with a 2960 switch and two PCs.

Device inventory: 4 × Cisco 2811 routers (each with a WIC-2T serial module), 4 × Cisco 2960 switches, and 8 × PC end hosts (two per LAN).

4 IP Addressing Scheme

4.1 Serial (WAN) Links — /30 Point-to-Point

Subnet	Local End	IP	Remote End	IP
10.0.12.0/30	R1 S0/0/0	10.0.12.1	R2 S0/0/1	10.0.12.2
10.0.23.0/30	R2 S0/0/0	10.0.23.1	R3 S0/1/0	10.0.23.2
10.0.34.0/30	R3 S0/1/1	10.0.34.1	R4 S0/0/1	10.0.34.2
10.0.41.0/30	R4 S0/0/0	10.0.41.1	R1 S0/0/1	10.0.41.2

Table 1. Serial link addressing (10.0.x.0/30 ranges).

4.2 LAN Segments and Loopbacks

Router	Gateway (Fa0/0)	Switch	Host Addresses	Router-ID (Lo0)
R1	192.168.1.1/24	Switch-1	PC1 .10 / PC2 .11	1.1.1.1
R2	192.168.2.1/24	Switch-2	PC3 .10 / PC4 .11	2.2.2.2
R3	192.168.3.1/24	Switch-3	PC5 .10 / PC6 .11	3.3.3.3
R4	192.168.4.1/24	Switch-4	PC7 .10 / PC8 .11	4.4.4.4

Table 2. LAN gateways, host addressing, and loopback router-IDs.

4.3 DCE / DTE Assignment

On each point-to-point serial link, one router acts as DCE (providing clocking via clock rate 64000) and the other as DTE. Because WIC-2T modules were installed in different slots, the DCE interface names differ across routers; the DCE end of each link is listed below.

Link	DCE End (clock rate)	DTE End	Subnet
R1 ↔ R2	R1 S0/0/0	R2 S0/0/1	10.0.12.0/30
R2 ↔ R3	R2 S0/0/0	R3 S0/1/0	10.0.23.0/30
R3 ↔ R4	R3 S0/1/1	R4 S0/0/1	10.0.34.0/30
R4 ↔ R1	R4 S0/0/0	R1 S0/0/1	10.0.41.0/30

Table 3. DCE/DTE assignment per serial link.

5 Configuration Methodology

Configuration proceeded in three phases: (1) physical build — placing devices, installing WIC-2T serial modules, and cabling the ring with serial DCE cables; (2) interface and OSPF configuration on each router; and (3) verification. A key practical finding was that the WIC-2T modules occupied different slots across routers, producing different serial interface names (Serial0/0/x on R1, R2, R4 versus Serial0/1/x on R3). Interface names were confirmed per device with show ip interface brief before addressing, and physical wiring was validated with show cdp neighbors to ensure both ends of every link shared the correct /30 subnet.

OSPF design choices: all four routers run a single process (router ospf 1) in Area 0; router-IDs are pinned to loopback addresses for stability; LAN interfaces are declared passive to suppress unnecessary hello traffic; and network statements use wildcard masks to enable OSPF precisely on the intended interfaces. Interface cost is left at the default (64 for the 64 kbps serial links, 1 for FastEthernet), which yields predictable, symmetric costs around the ring.

6 Verification and Results

The following screenshots were captured from the live topology and confirm the network is fully operational.

6.1 OSPF Neighbor Adjacencies

Command: **show ip ospf neighbor** (executed on R1)

```

249856K bytes of ATA System CompactFlash 0 (Read/Write)
Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
03:09:28: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial0/0/1 from LOADING to FULL, Loading Done
03:09:28: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on Serial0/0/0 from LOADING to FULL, Loading Done

R1>show ip ospf neighbor

Neighbor ID    Pri  State           Dead Time   Address      Interface
4.4.4.4        0    FULL/-         00:00:35    10.0.41.1    Serial0/0/0
2.2.2.2        0    FULL/-         00:00:35    10.0.12.2    Serial0/0/1
R1>show ip route ospf

 2.0.0.0/32 is subnetted, 1 subnets
O       2.2.2.2 [110/65] via 10.0.12.2, 00:02:34, Serial0/0/1
 3.0.0.0/32 is subnetted, 1 subnets
O       3.3.3.3 [110/129] via 10.0.12.2, 00:02:24, Serial0/0/1
        [110/129] via 10.0.41.1, 00:02:24, Serial0/0/0
 4.0.0.0/32 is subnetted, 1 subnets
O       4.4.4.4 [110/65] via 10.0.41.1, 00:02:34, Serial0/0/0
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.0.23.0 [110/128] via 10.0.12.2, 00:02:34, Serial0/0/1
O       10.0.34.0 [110/128] via 10.0.41.1, 00:02:34, Serial0/0/0
O       192.168.2.0 [110/65] via 10.0.12.2, 00:02:34, Serial0/0/1
O       192.168.3.0 [110/129] via 10.0.12.2, 00:02:24, Serial0/0/1

```

Figure 3. R1 shows two neighbors — 2.2.2.2 (R2) and 4.4.4.4 (R4) — both in the FULL state, confirming adjacency around the ring.

Each router forms a FULL adjacency with both of its ring neighbors, confirming that hello parameters, subnet masks, and authentication keys match on every link.

6.2 OSPF Routing Table

Command: **show ip route ospf** (executed on R1)

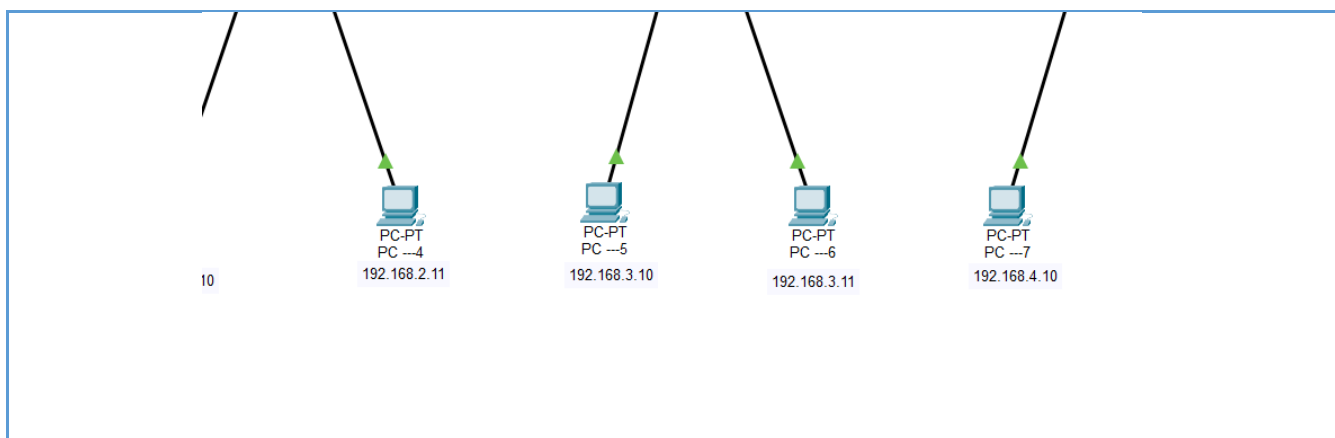


Figure 4. R1's OSPF routing table — 'O' routes to all remote LANs and loopbacks, with equal-cost dual paths to 3.3.3.3 and 192.168.3.0 (the ring's redundancy).

The two equal-cost entries for 192.168.3.0 and 3.3.3.3 (both cost 129, reachable via 10.0.12.2 and 10.0.41.1) are direct evidence of the ring's redundancy: OSPF computed a path in each direction. The differing costs — 65 to one-hop LANs versus 129 to two-hop destinations — confirm cumulative cost-based path selection.

6.3 End-to-End Connectivity Test

Command: **ping 192.168.3.10** (from PC1 on R1's LAN to PC5 on R3's LAN)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.10: bytes=32 time=8ms TTL=125
Reply from 192.168.3.10: bytes=32 time=10ms TTL=125
Reply from 192.168.3.10: bytes=32 time=10ms TTL=125

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 10ms, Average = 9ms

C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time=14ms TTL=125
Reply from 192.168.3.10: bytes=32 time=4ms TTL=125
Reply from 192.168.3.10: bytes=32 time=4ms TTL=125
Reply from 192.168.3.10: bytes=32 time=14ms TTL=125

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 14ms, Average = 9ms

C:\>
```

Figure 5. Successful ping from PC1 (192.168.1.10) to PC5 (192.168.3.10) across the ring — Sent = 4, Received = 4, 0% loss.

A successful ping between hosts on non-adjacent LANs proves OSPF distributed reachability correctly and that traffic is routed end-to-end across multiple routers.

6.4 Path Trace

Command: **tracert 192.168.3.10** (from PC1)

```
C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.10: bytes=32 time=8ms TTL=125
Reply from 192.168.3.10: bytes=32 time=10ms TTL=125
Reply from 192.168.3.10: bytes=32 time=10ms TTL=125

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 10ms, Average = 9ms

C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time=14ms TTL=125
Reply from 192.168.3.10: bytes=32 time=4ms TTL=125
Reply from 192.168.3.10: bytes=32 time=4ms TTL=125
Reply from 192.168.3.10: bytes=32 time=14ms TTL=125

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 14ms, Average = 9ms

C:\>tracert 192.168.3.10

Tracing route to 192.168.3.10 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    192.168.1.1
  1  2 ms    3 ms    7 ms    10.0.12.2
  2  8 ms    1 ms    1 ms    10.0.23.2
  3  4 ms    0 ms    6 ms    192.168.3.10

Trace complete.

C:\>
```

Figure 6. Traceroute from PC1 to PC5: 192.168.1.1 (R1) → 10.0.12.2 (R2) → 10.0.23.2 (R3) → 192.168.3.10 (PC5).

The trace confirms the exact router-by-router path packets take across the OSPF ring.

7 Security Hardening — OSPF MD5 Authentication

Threat. By default, OSPF trusts any device on a connected link that can produce valid hello and Link-State Advertisement (LSA) packets. An attacker with access to a serial link (or an emulated router) could inject forged LSAs advertising false routes, redirecting traffic through an attacker-controlled path (man-in-the-middle) or blackholing it. This is a control-plane attack that undermines the integrity of the entire routing domain.

Mitigation. MD5 message-digest authentication was enabled on every serial interface of all four routers. Each router computes a keyed hash over its OSPF packets; a neighbor accepts an update only if it reproduces the same digest using the shared secret key. Packets that fail — including any injected without the key — are silently discarded.

Configuration applied to each serial interface (example, R1):

```
interface Serial0/0/0
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 Cisco123
!
interface Serial0/0/1
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 Cisco123
```

Verification. After matching keys were applied on both ends of every link, show ip ospf interface reports authentication enabled and adjacencies remain FULL — confirming legitimate neighbors still communicate while unauthenticated routers would be rejected.

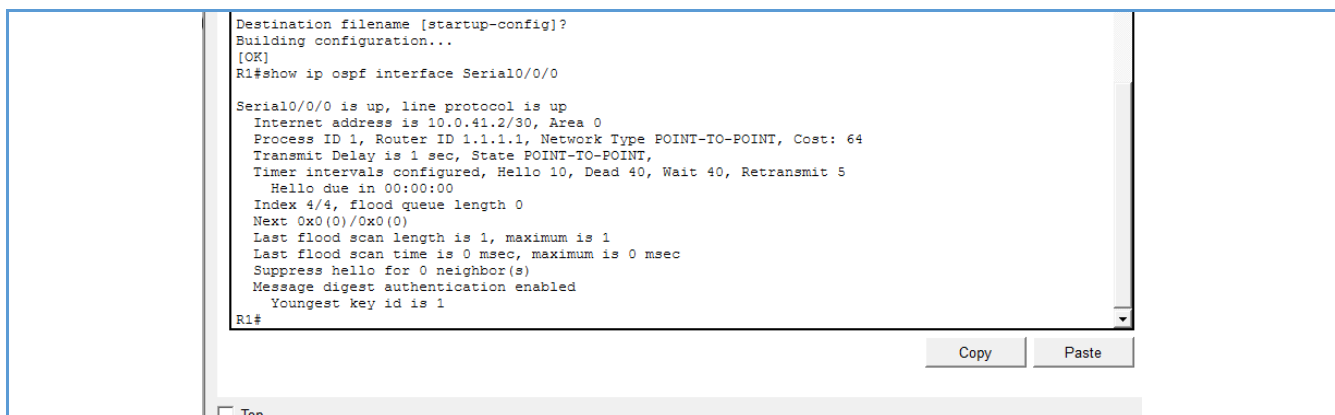


Figure 7. 'show ip ospf interface Serial0/0/0' on R1 reporting 'Message digest authentication enabled, Youngest key id is 1' on an interface that remains up.

8 Challenges and Resolutions

Several realistic issues arose during the build and were resolved systematically:

- **Inconsistent serial interface names.** WIC-2T modules landed in different slots, so serial ports were Serial0/0/x on some routers and Serial0/1/x on others. Resolved by confirming names per device with show ip interface brief before configuring.
- **Subnet mismatches on links.** Because cables were connected across several attempts, some serial IPs did not match the physically wired port. Resolved by mapping the real wiring with show cdp neighbors and re-aligning each link's /30 addresses.
- **Overlapping-address errors during correction.** Swapping two interface IPs directly triggered '% overlaps' errors; resolved by clearing one interface (no ip address) before reassigning both.
- **Missing ring-closing link.** One serial cable had not connected to the intended ports, splitting the topology into two islands. Identified via CDP (a neighbor was absent) and fixed by re-drawing the serial DCE cable between the correct ports.

9 Conclusion

The four-router OSPF ring was successfully designed, built, verified, and hardened. All adjacencies reached the FULL state, every router learned the complete network via OSPF, and end-to-end connectivity across the ring was confirmed with a zero-loss ping and a traceroute. The topology's redundancy was evidenced by equal-cost dual paths in the routing table. Finally, MD5 authentication was applied and verified on all serial links, addressing a well-known OSPF security weakness and aligning the exercise with Certified Ethical Hacker defensive objectives.

A Appendix A — Full Router Configurations

Configurations include the MD5 authentication applied in Section 7. R3's serial ports are Serial0/1/0 and Serial0/1/1 (WIC-2T in slot 1); the DCE end of each link (carrying clock rate 64000) is identified in Table 3.

A.1 Router R1

```
hostname R1
no ip domain-lookup
!
interface Loopback0
 ip address 1.1.1.1 255.255.255.255
!
interface FastEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
!
interface Serial0/0/0
 description Link to R2 (DCE)
 ip address 10.0.12.1 255.255.255.252
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 Cisco123
 clock rate 64000
 no shutdown
!
interface Serial0/0/1
 description Link to R4 (DTE)
 ip address 10.0.41.2 255.255.255.252
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 Cisco123
 no shutdown
!
router ospf 1
 router-id 1.1.1.1
 passive-interface FastEthernet0/0
 network 192.168.1.0 0.0.0.255 area 0
 network 10.0.12.0 0.0.0.3 area 0
 network 10.0.41.0 0.0.0.3 area 0
 network 1.1.1.1 0.0.0.0 area 0
!
end
```

A.2 Router R2

```
hostname R2
no ip domain-lookup
!
interface Loopback0
 ip address 2.2.2.2 255.255.255.255
!
interface FastEthernet0/0
 ip address 192.168.2.1 255.255.255.0
 no shutdown
!
interface Serial0/0/0
 description Link to R3 (DCE)
 ip address 10.0.23.1 255.255.255.252
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 Cisco123
 clock rate 64000
 no shutdown
!
interface Serial0/0/1
 description Link to R1 (DTE)
 ip address 10.0.12.2 255.255.255.252
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 Cisco123
 no shutdown
!
router ospf 1
 router-id 2.2.2.2
 passive-interface FastEthernet0/0
 network 192.168.2.0 0.0.0.255 area 0
 network 10.0.23.0 0.0.0.3 area 0
 network 10.0.12.0 0.0.0.3 area 0
 network 2.2.2.2 0.0.0.0 area 0
!
end
```

A.3 Router R3

```
hostname R3
no ip domain-lookup
!
interface Loopback0
 ip address 3.3.3.3 255.255.255.255
!
interface FastEthernet0/0
```

```
ip address 192.168.3.1 255.255.255.0
no shutdown
!
interface Serial0/1/1
description Link to R4 (DCE)
ip address 10.0.34.1 255.255.255.252
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 Cisco123
clock rate 64000
no shutdown
!
interface Serial0/1/0
description Link to R2 (DTE)
ip address 10.0.23.2 255.255.255.252
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 Cisco123
no shutdown
!
router ospf 1
router-id 3.3.3.3
passive-interface FastEthernet0/0
network 192.168.3.0 0.0.0.255 area 0
network 10.0.34.0 0.0.0.3 area 0
network 10.0.23.0 0.0.0.3 area 0
network 3.3.3.3 0.0.0.0 area 0
!
end
```

A.4 Router R4

```
hostname R4
no ip domain-lookup
!
interface Loopback0
ip address 4.4.4.4 255.255.255.255
!
interface FastEthernet0/0
ip address 192.168.4.1 255.255.255.0
no shutdown
!
interface Serial0/0/0
description Link to R1 (DCE)
ip address 10.0.41.1 255.255.255.252
ip ospf authentication message-digest
```

```
ip ospf message-digest-key 1 md5 Cisco123
clock rate 64000
no shutdown
!
interface Serial0/0/1
description Link to R3 (DTE)
ip address 10.0.34.2 255.255.255.252
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 Cisco123
no shutdown
!
router ospf 1
router-id 4.4.4.4
passive-interface FastEthernet0/0
network 192.168.4.0 0.0.0.255 area 0
network 10.0.41.0 0.0.0.3 area 0
network 10.0.34.0 0.0.0.3 area 0
network 4.4.4.4 0.0.0.0 area 0
!
end
```